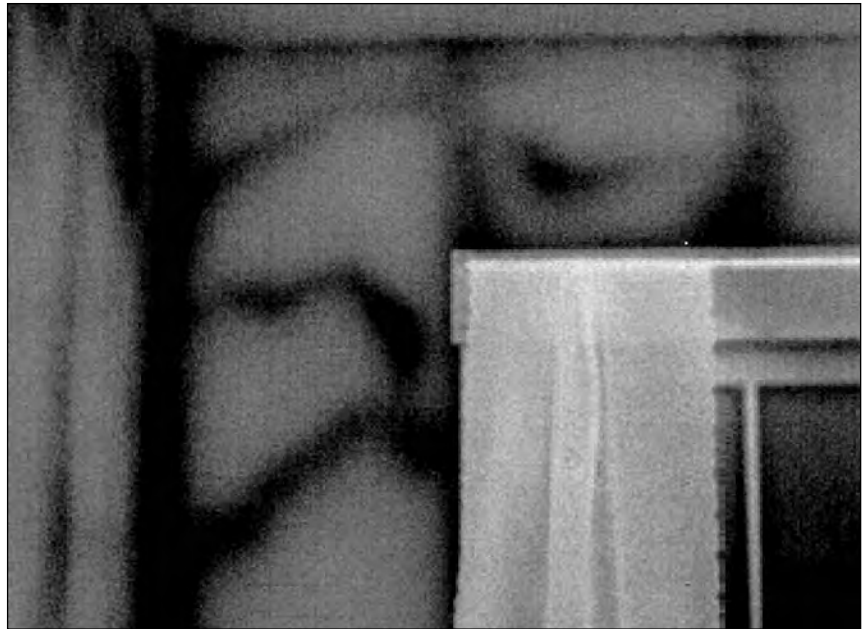


The Latest in Hot Shots

Infrared cameras are becoming affordable tools for building performance professionals.

by John Snell



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Infrared thermography is a proven tool that has been used for building diagnostics for a number of years (see “Selecting an Infrared Imaging System,” *HE* July/Aug ’93, p. 37). There is simply no better way to locate and diagnose problems with insulation performance and air leakage sites. You’ll not find a better tool for troubleshooting problems or prejob planning; the list of applications is long.

Infrared (IR) systems with focal plane array (FPA) detectors, first introduced in the 1990s, have evolved fully, and second-generation products are now reaching the market. Today’s IR systems no longer require cooling as did older technologies, and they are small and easy to use.

Unfortunately, in the past thermography—especially the high-end systems—was too expensive for many of us. But now things are finally changing for building contractors, weatherization professionals, and building scientists who want to use IR technology. New markets—in particular firefighting, law enforcement, and navigational aids—are causing manufacturers to produce better, smaller systems at lower cost. While high-

end systems can still push \$70,000, we now have a choice of great systems for less than \$25,000. Soon you’ll see very adequate systems for \$10,000, with a few simpler ones priced as low as \$5,000.

What Features Are Important?

The ideal infrared camera for today’s building contractor does not necessarily need to be top of the line. Many of the lower-cost systems are more than adequate in terms of thermal sensitivity, spatial resolution, and digital storage capabilities; they are also lightweight and powered by long-lasting batteries. You can add optional color imagery and radiometric temperature measurement. While neither is essential, both are features that you will frequently find useful. What other features should you look for in a system?

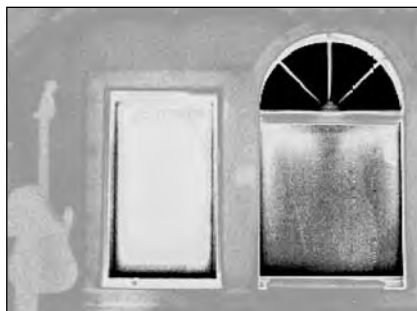
Thermal sensitivity. Thermal sensitivity is a measure of the minimum temperature difference that can be detected by an IR camera. Thermal sensitivity is no longer a big issue, because nearly every camera on the market today is more than adequate. Most cameras can

resolve a temperature difference of 0.2°C (0.4°F) or better at 30°C (86°F).

Field of view. A lens with a fairly wide field of view (FOV) is recommended, so you can use it both inside and outside the building. The specification is given in degrees for both horizontal and vertical views. A system with an FOV of approximately 20° x 20° or greater will perform well. Spatial resolution or instantaneous field of view (IFOV) is based on the lens being used as well as the infrared detector. If spatial resolution is too low, the image will appear fuzzy. An IFOV of 2.6 milli-Radians or greater is considered acceptable.

Detector matrix. A related specification for FPA systems is the size of the detector matrix. A 320 x 240 detector has over 76,000 pixels while a 160 x 120 detector has one-quarter that number. Given currently available image-processing routines, 160 x 120 is considered the minimum number of pixels needed for good imaging. It’s always a good idea to try before you buy.

Dynamic range. A dynamic range allowing for storage of 12-bit or 14-bit images (which can be manipulated after capture) is preferred. Several systems



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The thermal performance of windows during the heating season becomes visible based on their infrared signatures. The top window on the right is single-glazed; the window below is double-glazed; and the one on the left is argon-filled, double-glazed.



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Modern hand-held infrared systems are easy to use; the optional LCD display makes it easy to share information with others standing nearby.

store data only as an 8-bit file; if mistakes are made during data capture, they cannot be corrected later. Certainly an 8-bit system is usable, but the operator must understand how to work within the system's inherent limitations.

Image storage and retrieval. Digital image storage with optional analog video output is the norm. Data are typically stored on a PC card or in electronic media. A recently introduced system uses a removable personal digital assistant (PDA) for image viewing and data storage. Reports are typically created in proprietary software, some of which can be expensive. Options may be available to download a nonproprietary image into a word processing document. Many IR systems now sport large liquid crystal display (LCD) screens—either built in or available as an option. This highly recommended feature makes it easy to show the thermal image to coworkers, contractors, or owners.

Table 1. Imaging Technologies

Technology	Brands (under \$25,000)
Ferroelectric FPA	Raytheon PalmIR Pro and Palm 225D MIT Heat-FindIR* • ISI VideoTherm 2000*
Ferroelectric, 2 nd generation	Raytheon PalmIR 500-D • Land Cyclops PPM Plus
Microbolometer FPA	FLIR 515 • Mikron 7515 • Indigo ThermoCorder
Scanned-array FPA	Infrared Solutions SnapShot • Iacon digiCamIR
Amorphous-silicon FPA	Coming soon...

* These cameras are no longer in production.

Radiometric temperatures. Radiometric systems measure radiant energy and convert that information into an apparent temperature value. Being able to measure radiometric temperatures can be very useful for building professionals, but it is by no means essential. In the past, radiometric systems were very expensive; that's not necessarily true anymore. Several camera systems that are less than \$25,000 are fully radiometric, meaning that the entire image is calibrated for temperature measurement. Others have a single spot of temperature measurement in the center of the image. In either case, having the capacity to measure and show temperatures on the image is a great plus.

With all of the features available, it is easy to become infatuated with a new IR system and forget about your budget. Some manufacturers have cleverly priced their systems to make it easy to add on a long list of features—for a price, of course. Make sure you look at the total package price, including lenses, reporting and/or analysis software, data storage cards, LCD view screens, batteries, and training.

Although a new IR system will probably not often need service, when it does, you want to make sure that the work is done quickly and effectively. Before you buy a camera, ask about repairs. What is the average turnaround time? Are free loaners available if repairs take longer than expected? How much should you budget annually for upkeep? The best system in the world is of little value if it is in for repairs when you really need it.

Most IR system users budget a very conservative 5%–10% of the cost of the system for annual repairs and maintenance.

Annual calibration of IR systems is not required, despite the fact that some manufacturers push it. Calibration checks are recommended before and after each use. Checking the temperature of a person's tear duct area, which should be between 94°F and 97°F, will tell you if your system is calibrated properly. If you get another measurement, calibration by the manufacturer may be necessary.

Anyone who can use a conventional video camcorder can probably quickly learn to get good thermal images. Be aware, however, that the accurate interpretation of the images requires not only good camera skills, but also an understanding of heat transfer, radiation physics, and basic building diagnostics. Without that foundation, expect to make several serious mistakes. Some training options are available, but make sure the one you choose covers your camera and building applications. For groups of five to ten people, specialized, on-site training focused on their needs and equipment probably makes the most sense.

IR Camera Technologies

Currently, three technologies are being used to manufacture camera systems priced at less than \$25,000 (see Table 1).

Ferroelectric. A widely used technology utilizes a ferroelectric FPA detector manufactured by Raytheon Infrared. This



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The Raytheon Palm 500D, also sold as the Land Cyclops PPM Plus, incorporates a PDA as the display and image storage device.



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The Infrared Solutions' SnapShot sports a large LCD screen.

is a thermal detector that responds to radiant energy with a resistance change. This commercialized military sensor first found its way into commercial IR cameras used by firefighters, as well as on the front end of the Cadillac deVille as a navigational aid. (Expect to see more navigational aids in other vehicles, both cars and large trucks, in the future, as these aids dramatically assist vision at night and during inclement weather. In the Cadillac, the thermal image is projected onto the windshield much like a heads-up display.)

The Raytheon PalmIR Pro is currently available and is appropriate for building inspections. Two other systems employing ferroelectric detectors, the ISIVideoTherm 2000 and the MIT Heat-FindIR, are no longer in production, but both are still supported if service is required.

One side effect of the ferroelectric detector is that images sometimes retain



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Several cellulose voids are visible in this house under the windows. The diffuse pattern on the left is caused by a warm chimney built into the outside wall.

thermal artifacts or halos. This is most noticeable around the edges of objects having high thermal contrast, and is most pronounced when the image is poorly adjusted. Until recently it has also meant that radiometric measurements were simply not possible. Second-generation systems have corrected this problem. Image quality is dramatically improved and the halo problems have been eliminated; in addition there is now a spot measurement at the center of the image. The Raytheon PalmIR 500-D (also marketed as the Land Cyclops PPM Plus) is the first of these systems on the market; interestingly, the design cleverly incorporates a PDA as the view screen and image storage device.

Microbolometer. A microbolometer is an FPA with a thermal detector that responds to thermal radiation with a voltage change. These systems typically produce high-quality images, but, until recently, for a fairly high price. Responding to market demand, several new systems have been introduced in the under-\$25K category and others will surely follow. The Mikron 7515 offers 320 x 240 resolution with single-point temperature measurement in the image. FLIR has begun offering a microbolometer with nonradiometric 160 x 120 resolution and 8-bit image storage (FLIR 515).

Another type of microbolometer that warrants consideration is represented by

Infrared Solution's SnapShot and Ircon's digi-CamIR. These systems are 160 x 120 FPAs but they are also fully radiometric. The images are displayed on a large LCD screen and captured on a PC card. The image is produced using an array scanned over a 1.6-second period; this makes it very difficult to take images of moving objects or processes. The system can be used to scan slowly over a wall. However, a better mode of operation is the snapshot mode (one image at a time). The price

tag of approximately \$13K has made these systems an attractive alternative in the market.

Another microbolometer that bears mention is the Indigo ThermoCorder. The infrared camera of this infrared/visual-recording device is tiny (1.6 inches x 1.6 inches x 3 inches). Although the black-and-white image has a pixel density of 160 x 128, the image is more than adequate for building work. The really cool trick, however, is that Indigo has married its miniature IR camera to a Sony digital camcorder. With the flip of a switch you can record either IR or visual images directly into the Sony, either as a single frame or as a burst of digital video.

Amorphous silicon. The newest technology, amorphous-silicon (a-Si) detectors are made using conventional manufacturing processes, less costly base materials, and off-the-shelf electronics; as a result the cost of a-Si-based systems is significantly lower than that of previous technologies. The first systems are beginning to show up in the law enforcement and firefighting markets. Other commercial systems should be available soon.

Can Your Company Afford a Camera?

Consider the fact that a new IR system will last ten years or more. Even if you budget 10% annually for maintenance, the cost for the \$25,000

equipment is \$2,750 per year over its minimal life. If the system is used only 100 times each year—a low figure—the cost for the infrared equipment is only \$27 per inspection. Even with other associated costs, the total cost of a production inspection is probably less than \$100 per home.

If you are a contractor, doing an infrared preinspection of a job can mean that things go smoothly and on budget, because you will quickly understand exactly how the building is constructed and how it is performing. And if you do an inspection after the job is complete, you will virtually eliminate costly callbacks. In either case, the cost is small over the life of the energy measure.



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